An Intrusion Detection Method for Automotive Ethernet Networks Based on Time Series Prediction and Rule Setting

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***Abstract*—Nowadays, intelligent and connected vehicles (ICVs) have rapid development, and their cyber security problems have become increasingly significant. The cyber-attacks on ICVs directly threaten the property, privacy and even personal safety of drivers and passengers. Most existing intrusion detection methods are primarily designed for CAN bus, with relatively little research focused on automotive Ethernet. This paper takes SOME/IP as an example and focuses on attack detection in automotive Ethernet. An intrusion detection method combining time series prediction and rule-based approaches is proposed. An in-vehicle network traffic prediction model that combines CNN-BiLSTM with attention mechanisms is utilized to learn the temporal characteristics of data packets in automotive Ethernet. The cosine similarity between real messages and predicted results is utilized as a rule to identify attacks. Experimental results demonstrate the effectiveness of the proposed method.**

Keywords—intelligent and connected vehicles, intrusion detection, time series prediction

I. Introduction

In recent years, the modern automobile industry has been developing rapidly in the direction of intelligence and connectivity. It brings great change to the transportation system and people's travel. Advanced intelligence technologies in various fields such as intelligent sensors, intelligent algorithms, and artificial intelligence are widely used in the perception, decision-making and control process of modern vehicles, which makes ICVs a complex and intelligent system. The number of ECUs in the vehicle has ushered in explosive growth. For example, some luxury models even have more than 100 ECUs [1]. The rapid increase in the number of ECUs not only leads to an increase in wiring harness costs and assembly costs but also makes it impossible for vehicles to complete rapid iterations like other IoT devices or smartphones to keep up with the pace of information technology development. However, this problem can be solved by a domain-centralized electrical/electronic architecture (EEA) and a vehicle-centralized EEA, which build a new in-vehicle communication network with in-vehicle Ethernet as the core backbone network. The Ethernet network-based vehicular EEA could reduce the cost of automobile manufacturing and accelerate the iteration of automobile software and hardware. The in-vehicle Ethernet could also provide higher communication bandwidth for communication traffic required by V2X, human-machine interaction required by in-vehicle infotainment systems, and some sensors for intelligent driving, such as cameras and lidar [2]. In-vehicle Ethernet contains numerous protocol clusters. Among them, SOME/IP (Scalable Service-Oriented Middleware over IP) is a very promising application layer protocol, and it is gradually popularized in in-vehicle networks [3-5].

The development of vehicle connectivity brings many positive changes to the traffic environment and travel experience, and also leads to the issue of vehicle cyber security. In the past decade, the number of automotive cyber-attacks has risen sharply. According to Upstream Security, it has increased sixfold from 2016 to 2019, doubling from 2018 to 2019 [6, 7]. In addition, more than half of these cyber-attacks were made by hackers, not security researchers: 57% and 55% of automotive cyber security incidents were initiated by cybercriminals in 2019 and 2020, respectively [7, 8]. Among them, the remote attack incident of Jeep Cherokee in 2015 is widely known in the industry. Charlie Miller and Chris Valasek successfully hacked a Jeep Cherokee by a remote attack and interfered with its steering and braking functions, which eventually led to the emergency recall of 1.4 million related models of Chrysler [9, 10]. In a sense, a new chapter in automotive cyber security was

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opened by this incident. Since then, governments and manufacturers have paid more and more attention to automotive cyber security, and automotive cybersecurity has greatly expanded significantly from academia to industry.

Intrusion detection system (IDS) serves as a critical measure in ensuring automotive cybersecurity. In-vehicle network intrusion detection can proactively identify malicious attacks in real-time. However, the application of intrusion detection in automotive Ethernet networks encounters challenges. Firstly, the detection methods need to be adapted to the in-vehicle Ethernet protocol. Some attacks on automotive Ethernet may be more complex than CAN bus, making them difficult to detect solely from a statistical analysis perspective. It might be necessary to utilize temporal characteristics in the context to detect these stealth attacks. Moreover, as automotive Ethernet has not yet been widely adopted, there is a lack of datasets in this area to support research, especially attack samples.

To address the above issues, this paper focuses on the new in-vehicle communication network with Ethernet as the backbone and proposes a deep learning and rule combined-based IDS to detect the intrusion of the SOME/IP communication network. Two SOME/IP datasets including normal packets and six types of intrusion packets are generated and labelled respectively which are used to test and evaluate the performance of the designed IDS.

The main contributions are summarized as follows:

(1) A novel intrusion detection method combining deep learning traffic prediction and rules is constructed to recognize attacks.

(2) This paper focuses on intrusion detection for automotive Ethernet. A dataset of the SOME/IP protocol is generated and used for training and testing the proposed intrusion detection method.

(3) Cosine similarity-based rules are designed to identify the attacks.

The rest of this paper is organized as follows: Section 2 reviews the related works in this field. Section 3 details the construction of the proposed algorithm. Section 4 conducts the generated dataset and experimental evaluation of the designed IDS. Section 5 summarizes the full text and draws conclusions.

II. RELATED WORKS

This section describes the existing in-vehicle network intrusion detection system, the communication method of SOME/IP protocol and the existing security protection method, and the time series prediction method. The features of SOME/IP protocol are considered when designing the in-vehicle Ethernet dataset. These time series prediction related works contribute to the design of the intrusion detection model in this paper.

A. In-Vehicle Network Intrusion Detection System

In-vehicle network intrusion detection system can be broadly categorized into rule-based, statistical-based, physical feature-based, and deep learning-based approaches.

The rule-based IDS detects deviant messages through a list of predefined attack signatures. Dagan et al. [11] proposed an anti-spoofing system that can detect counterfeit messages from the illegal ECU based on the rules about arbitration bits. The statistics-based method generally utilizes statistics that can be obtained from messages in the network layer. Song et al. [12] presented a lightweight IDS in which spoofing attacks can be quickly detected by a temporal interval of messages. The physical feature-based approaches work mainly on the physical layer of the network. Deng et al. [13] create a voltage fingerprint for each ECU to detect malicious attacks. At present, the deep learning method has been widely used in the construction of IDS for in-vehicle networks. Song et al. designed an IDS with a DCNN as the core algorithm [14]. Tariq et al. [15] proposed a LSTM-based intrusion detection method, named CANtransfer. Hanselmann et al. [16] presented an LSTM with auto-encoder IDS.

However, these IDSs are designed for in-vehicle CAN bus, and there are fewer studies on IDSs for in-vehicle Ethernet. This is mainly because there are few public datasets benchmark for in-vehicle Ethernet protocols such as SOME/IP compared to CAN datasets. The current in-vehicle network market is still dominated by distributed EEA and CAN bus, and the research on in-vehicle Ethernet security protection is relatively lagging behind. With the development of the vehicle EEA, the In-vehicle Ethernet, as the backbone network of domain-centralized and vehicle-centralized, is expected to dominate the in-vehicle network together with the CAN bus in the future. Therefore, it is necessary to research the intrusion detection of in-vehicle Ethernet.

B. Scalable Service-Oriented Middleware over IP

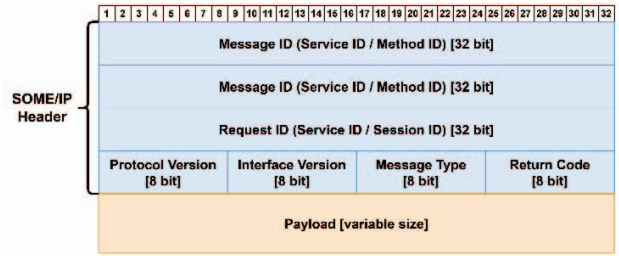

Figure 1. The frame format of SOME/IP

SOME/IP is a scalable middleware for transmitting service information. It can adapt to different sizes of devices based on different operating systems, from small cameras to large vehicle systems or autonomous driving modules. Compared with the signal-oriented communication method of the traditional CAN bus, SOME/IP is a service-oriented communication method, and its data frame format is shown in Figure 1. In this paper, some data in the SOME/IP data frame header are used as the input features of the designed IDS.

Up to now, three main communication models are provided by SOME/IP: Service Discovery (SD), Remote Procedure Calls (RPC) and Publish/Subscribe Mechanism. Among them, SD is mainly used to inform the availability and access methods of service instances, which is implemented through Find

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Service and Offer Service messages, as shown in Figure 2(a). Request/Response communication is one of the most common RPC communication models. It is implemented through Request and Response messages, as shown in Figure 2(b). The communication method is that the client sends a request, and the server processes and responds after receiving the request. Fire &Forge communication is another RPC communication model, which is implemented through the Request_no_return message, as shown in Figure 2(c). The communication method is that the client sends a request, and the server processes the request after receiving it but does not respond. Publish/Subscribe communication is mainly realized through Notification messages, and the communication flow is shown in Figure 2(d). The client subscribes to the required event group to the server, and if an event occurs, the server will send a notification to the client. In this paper, the hypothetical attacker has the ability to interfere with the normal communication model of SOME/IP, thereby sending fake messages in the communication network or blocking the delivery of real messages.

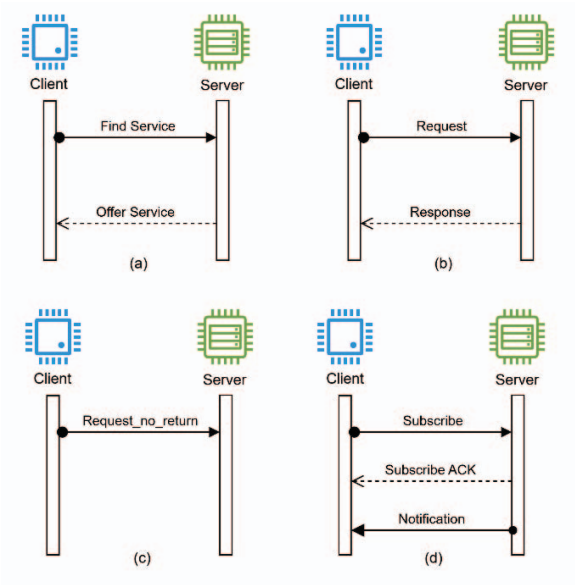

Figure 2. The main communication model of SOME/IP: (a) SD; (b) RPC1 - Request/Response; (c)PRC2 - Fire&Forget; (d) Publish/Subscribe Mechanism

In the field of in-vehicle Ethernet, there are few other researches on SOME/IP cyber security protection. Herold et al. proposed a SOME/IP protocol IDS named Esper, which detected attacks relying on a pre-compiled rule base[17]. The designed IDS was tested and verified by the aircraft cockpit communication network, and the results showed that the online detection performance of the IDS needs to be further optimized. Marco et al. completed cyber security protection for SOME/IP protocol based on applied cryptography in [18] and [19]. They designed a new security mechanism, adding headers to SOME/IP payload for authenticating requests and authenticating responses. However, the disadvantage is that the length of the payload that can be carried by a small number of packets is sacrificed in this method. Li et al. proposed a grey-box penetration testing tool named Ori[20]. SOME/IP penetration test packets with legitimate headers can be randomly generated by it to fuzz the SOME/IP communication network. Zelle et al. used the Tamarin formal analysis tool to formally analyze the cyber security performance of the SOME/IP protocol. The results showed that the protocol is highly vulnerable to vehicle hackers, such as man-in-the-middle attacks[21]. On this basis, they provided two different security mechanisms, SESO-RC and SESO-AS based on applied cryptography to provide confidentiality protection for SOME/IP communication. Based on applied cryptography technology, the author and his team also designed an efficient identity authentication and secure communication mechanism for the SOME/IP communication protocol and evaluated its security features and resource overhead through formal analysis and experimental analysis[22].

C. Time Series Prediction

Due to the complexity and stealthiness of some attacks on automotive Ethernet, temporal features that represent contextual relationships may be a viable method for detecting these attacks. The following is an overview of time series prediction methods.

Time series prediction relies on historical observation to construct models that forecast future trends in time series. Traditionally, this field has primarily leaned on statistical models such as Auto Regressive (AR), Moving Average (MA), Auto Regressive Moving Average (ARMA), and its extended version, ARIMA[23]. These methods capture linear dependencies within sequences, boasting mature theories and strong interpretability. However, when confronted with complex nonlinear time series data, the predictive performance of these methods often becomes constrained. In recent years, Recurrent Neural Network (RNN) and its variants, including Long Short-Term Memory (LSTM)[24] and Gated Recurrent Units (GRU)[25], have revolutionized the handling of sequential data by introducing a "memory" mechanism. These models effectively address the limitations of traditional methods, enabling them to capture long-term dependencies in time series. Additionally, Convolutional Neural Network (CNN)[26] and Transformer models[27] have also demonstrated potential in time series prediction, particularly shining in processing high-dimensional, multivariate time series data.

Given the intricate context-aware data in in-vehicle Ethernet, constructing time series prediction models that are more efficient, accurate, and interpretable remains a challenge.

III. APPROACH

This paper presents a two-part intrusion detection method that combines time series prediction with rules. The method consists of a CNN-BiLSTM with an attention model to predict vehicle network traffic and a cosine similarity difference rule-based intrusion identification model to detect attacks.

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A. Intrusion Detection Method Overview

The intrusion detection method proposed in this paper is used to monitor the intrusion against the SOME/IP communication network. This method primarily consists of two components: an in-vehicle network traffic prediction model and an intrusion identification model.

The in-vehicle network traffic prediction model is the core of the intrusion detection system, utilizing the time series prediction method. Considering the scarcity of attack samples in automotive Ethernet, it is impractical to rely on positive sample detection in real-world applications. Therefore, our time series prediction model will be trained on attack-free data, learning the contextual relationships within normal data. Another reason for adopting a time series model is that some attacks on automotive Ethernet are more complex. Unlike the CAN bus, automotive Ethernets have specific senders and receivers. Attackers targeting Ethernet often possess stronger knowledge and experience, so some attack packets may conform to the Ethernet protocol and match the original controller's transmission frequency. Thus, it becomes necessary to detect these through contextual relationships over a period of time.

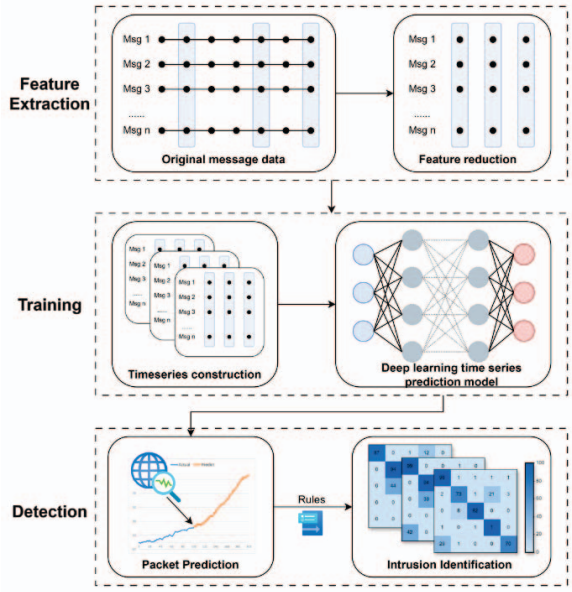

Figure 3. The overall process of the designed IDS

The intrusion identification model operates on the basis of network traffic predictions, using manually set rules or thresholds to differentiate between normal and intrusive traffic. The specific settings of these rules depend on the performance of the in-vehicle network traffic prediction model. It means that the prediction must conform to the characteristic distribution of normal data. This is essential to ensure that the established rules can effectively distinguish between normal and abnormal packets. However, this does not mean that it is necessary to aim for particularly high prediction accuracy. This is because it is difficult for the dataset to cover all vehicle operating conditions, and the need to improve the accuracy may require a lot of arithmetic power and training time to support it.

The working process of intrusion detection is divided into three main steps: feature extraction, network training, prediction and intrusion identification, as shown in Figure 3.

In the feature extraction stage, in order to improve the performance of the proposed deep learning model, the original message is used as the input, and the feature reduction method is applied. Only the feature information with high correlation is extracted. The output data of this stage is a two-dimensional feature matrix composed of several one-dimensional feature vectors. The network training stage is the second stage. The two-dimensional matrix is first reorganized into a three-dimensional time series tensor, which is then fed into a deep learning network based on CNN, BiLSTM and attention layers. The CNN, BiLSTM and attention layer could capture and learn potential spatiotemporal nonlinearity in the time series data, so as to enable the generation of packets that comply with features of the temporal relationships of SOME/IP normal data packet. The dimension of the output data in this stage is reduced and the data becomes a two-dimensional matrix composed of one-dimensional vectors. In the detection stage, some packets are fed into the in-vehicle network traffic prediction model. The model predicts the packets for the upcoming time period based on these inputs. The predicted packets are then compared with the actual packets inputted within a subsequent time window. The pre-set rules will identify whether it is an intrusion or not.

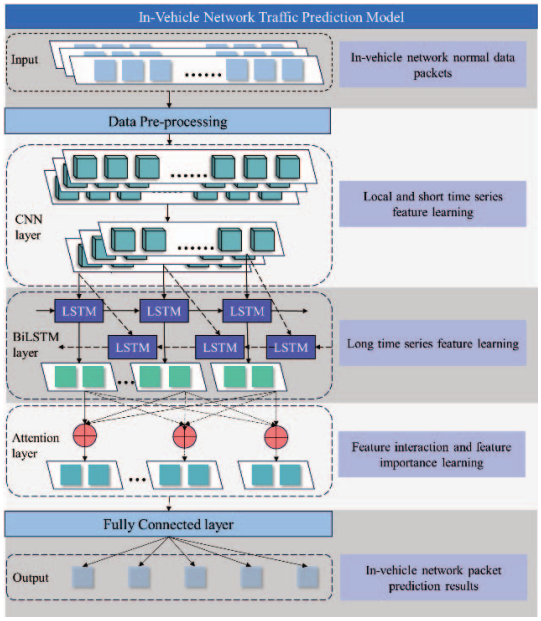

Figure 4. The architecture of in-vehicle network traffic prediction model

B. In-Vehicle Network Traffic Prediction Model

As previously mentioned, the in-vehicle network traffic prediction model is the core of the entire intrusion detection method. Therefore, to fully capture and learn the lateral relationships of features in automotive Ethernet packets, the short-term vertical time series relationships, long-term vertical time series relationships, and the importance of features, the CNN-BiLSTM with attention mechanism is employed. Figure

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4 shows the architecture of in-vehicle network traffic prediction model.

Convolutional operations can extract local relationships between adjacent data, both laterally and vertically. Given the relationships between successive packets and different features within packets in automotive Ethernet, a one-dimensional convolutional layer is established. This is followed by a max pooling layer, which aims to reduce data dimensions while preserving spatial invariance, thus lowering model complexity and computational demands. The pooling layer also helps prevent overfitting.

BiLSTM layers are capable of learning relationships between features that are distant in time series, and they can process both forward and backward time series data. However, due to the consideration of longer time series, they may not efficiently capture some local features. Therefore, in this study, two BiLSTM layers are employed after the convolutional layer. This combination leverages the strengths of both convolutional and LSTM layers, ensuring efficient learning of lateral relationships of features, as well as short-term and long-term vertical time series relationships in automotive Ethernet.

The attention mechanism is capable of learning the correlations between features, which serve as the basis for determining their importance. Integrating the attention mechanism allows for the weighting of learning parameters and the calculation of weight values. This mechanism assigns attention weights to the time series feature matrix extracted by the BiLSTM layer, thereby enhancing the impact of key features. Given that vehicle network communications adhere to specific rules and packets must conform to protocol standards, attacks on automotive networks typically target specific parts of the time series or specific positions within packets. The attention mechanism helps to identify these critical features, making it a valuable addition to identifying subtle anomalies in network traffic.

C. *Intrusion Identification Model*

In contrast to other time series data, IVNs data presents unique challenges due to the uncertain nature of automotive operating scenarios and the difficulty in accurately predicting in-vehicle human behavior. Therefore, directly comparing the results of the in-vehicle network traffic prediction model with actual in-vehicle Ethernet messages can lead to a significant number of false positives. Therefore, we developed a rule-based intrusion identification model.

The rules in our model consist of two types. The first type involves extracting key features from both the in-vehicle Ethernet data and the prediction data. It is crucial to ensure that normal messages and anomalous messages have distinguishable projections on these features. This determination requires experimental evaluation. The second type of rule is event-based. Simply comparing the features of actual traffic with predicted traffic still has limitations. The time window used in training and utilizing the Traffic Prediction Model is relatively small in the context of automotive event behavior. Therefore, certain longer-term correlated event patterns may not be captured. Additionally, if an attacker has knowledge of the vehicle's protocol, they can send a small number of malicious messages that have a minimal impact on the key features of in-vehicle Ethernet. This makes it challenging to detect such attacks.

We identified the cosine similarity within a specified time window as the first type of rule. The second type of rule involves establishing associations between in-vehicle Ethernet messages and specific events such as in-vehicle infotainment and advanced driver assistance. Typically, this knowledge is proprietary information held by automotive manufacturers. Reverse engineering of the Ethernet is required to establish these associations. Therefore, in the subsequent experiments of this paper, only the first type of rule will be considered.

IV. EXPERIMENTAL

A. *SOME/IP Datasets Generation*

SOME/IP Generator [28] is used to construct a dataset containing normal and abnormal packets. The tool is implemented based on Python 3. It can generate normal packets according to a given communication network and generate attack packets according to the settings. Specifically, in the first step, the SOME/IP Generator tool, based on the settings, generates a pcap file containing all the message information. Then Python's CSV and scapy packages are used with Wireshark software to extract message information one by one.

Finally, the distribution of the generated datasets is shown in Table 1. Two datasets are constructed for training and testing the deep learning network, respectively. The training dataset contains 21287 normal packets and 5584 abnormal packets. The test dataset contains 21216 normal packets and 4948 abnormal packets.

Table 1. Training and testing datasets

Class	Anomaly type	Label	Training Dataset	Testing Dataset
Normal	-	1	21287	21180
Anomaly	Fake client ID	2	730	612
	Wrong interface	3	715	605
	Disturb Timing	4	582	632
	Fake response	5	1419	1175
	Send error on event	6	78	68
	DoS	7	2060	1920
Total	-	-	26871	26192

Six types of abnormal packets are generated, which are shown as follows.

(1) Fake client ID: Attack that sends an arbitrary SOME/IP Packet using the own configuration (IP and MAC) but impersonates another valid client id.

(2) Wrong interface: Attack that sends a valid message impersonating another device with the wrong interface Number.

(3) Disturb Timing: Selects a device that sends time-sensitive messages from configuration and sends an additional arbitrary but corrects SOME/IP Packet to disturb the timing of other devices.

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(4) Fake response: Attackers interfere with legitimate nodes by sending unsolicited response packets.

(5) Send error on event: According to the requirements of the AUTOSAR standard specification, a Notification type message should not be responded to with an Error type message. Attackers can break this limit, thereby interfering with the normal communication of the communication network.

(6) DoS: In order to consume the computing resources of other legitimate nodes and interfere with their normal business, attackers send a large number of valid packets to the communication network in a short period of time as an illegal node or after completely controlling a legitimate node.

Table 2. Raw data features and preprocessing

Feature	Description	Preprocessing
Number	Message number	Delete
Time	Message timestamp	-
Src MAC	MAC address of the sender (Source MAC address)	Delete
Dst MAC	MAC address of the receiver (Destination MAC address)	Delete
Src IP	IP address of the sender (Source IP address)	Encode
Dst IP	IP address of the receiver (Destination IP address)	Encode
Service ID	A unique ID for each service in the SOME/IP communication network	Encode
Method ID	A unique ID that indicates that a method belongs to a Service	Encode
Length	Payload length	-
Client ID	A unique ID for each client in the SOME/IP communication network	Encode
Session ID	Used to indicate the number of times the same client has requested the same service multiple times	Delete
Protocol version	Used to indicate the version number of the currently used SOME/IP protocol	Delete
Interface version	The major version number used to identify the service interface, defined by the user	Encode
Message type	Used to indicate different message types	Encode
Return code	Used to indicate whether the request is successfully processed, or to provide feedback on errors in the request	Encode
Periodicity	Used to indicate whether a service is executed periodically	Delete
Label	Label that indicates the classification of the message	Only use for test

B Data Preprocessing and Feature Extraction

Preprocessing and feature extraction on raw data is a critical stage in designing an efficient IDS. During this stage, the raw data is filtered to remove redundant features and incomplete data. Then, data that are difficult to directly recognize by the model, such as IP addresses, were encoded into numerical values. Additionally, the data in hexadecimal format were converted into decimal format.

Seventeen original features are contained in each message. The names and meanings of the features are shown in Table 2. After data preprocessing and feature extraction, the total number of features decreases to 10.

Some feature-provided information is redundant or invalid and needs to be removed to optimize the deep learning network training process. These features include Number, Src MAC, Dst MAC, Protocol version, and Session ID. Among them, the Number and Time provide the same timing information. The Src MAC and Src IP provide the same address information of the source. The Dst MAC and Dst IP provide the same address information of the destination. The protocol version and Session ID are not relevant for the intrusion detection of this paper.

We trained the in-vehicle network traffic prediction model using normal packets, which do not include the feature label. Feature labels are not involved during the model's prediction phase, but we compare the labels from the test set after obtaining results from the intrusion identification model. The rules we currently use are based on binary classification and are designed only to identify the presence of an attack, not the specific type. Therefore, we modified the labels in the test set to 0 or 1. It is important to note that setting identification rules for different types of attacks can specifically differentiate among attack types. However, considering the difficulty of acquiring real Ethernet attack samples in practical applications, our experiment employs a simpler binary classification rule.

C. Experiment Environment

The construction of the in-vehicle network time series prediction model and data processing were conducted in a Python 3.9 environment with TensorFlow 2.6.0. Training of the model was accelerated using an NVIDIA GeForce RTX 4070Ti with 12GB of memory.

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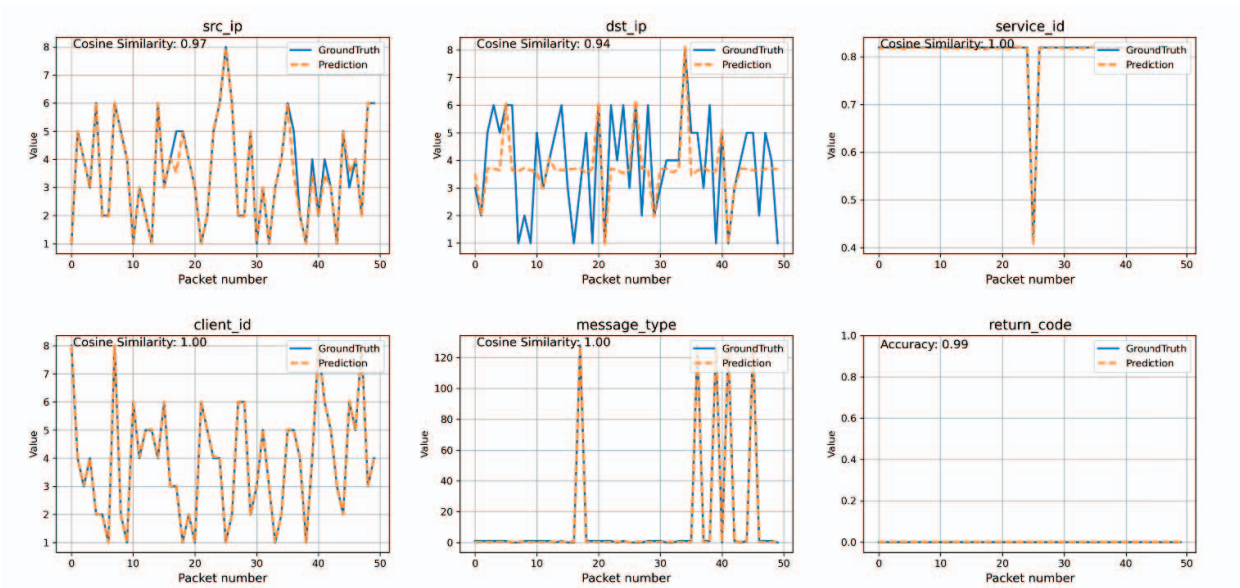

Figure 5. The predictive results under attack-free condition

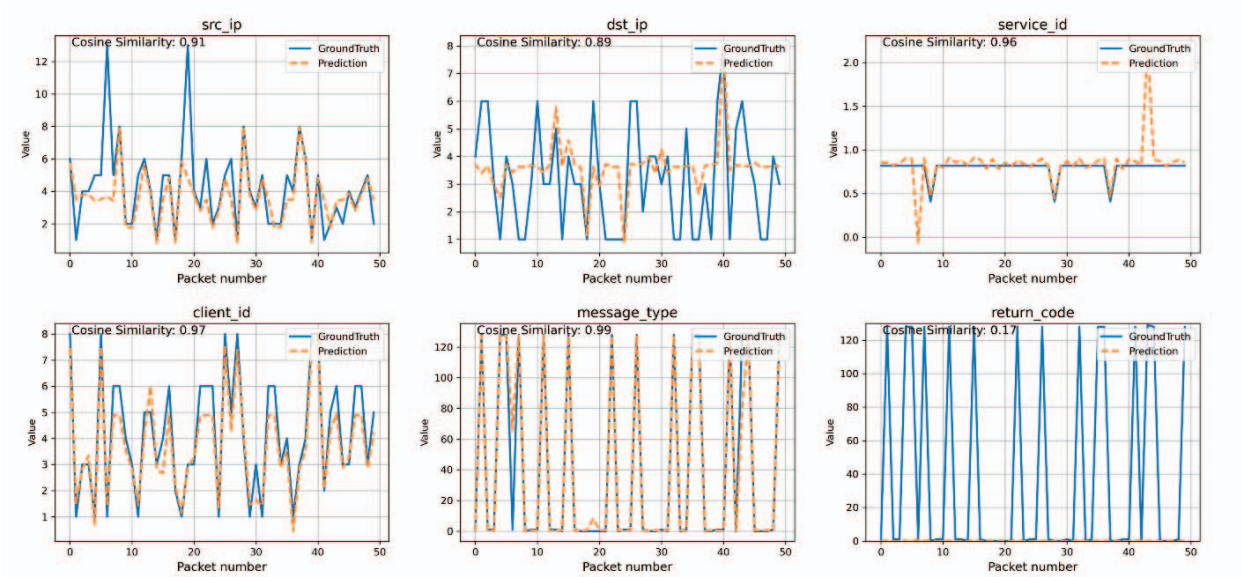

Figure 6. The predictive results under attack condition

D. *Experiment Results*

During the prediction process, we input the packets from the dataset into the in-vehicle network traffic prediction model in consecutive time windows and then aggregate the predicted results. Figure 5 shows the predictive results of 50 packets for several key features of SOME/IP packets under attack-free conditions. Figure 6 represents the predictive results under several types of attacks condition. The cosine similarity in Figure 5 and Figure 6 stands for the value of the 50 packets. It can be observed that the cosine similarity between the predicted results and the actual values is generally ⩾94%. This demonstrates that the in-vehicle network traffic prediction model has successfully learned the temporal relationships among SOME/IP packets. Figure 6 shows the parts of the predictive results for several key features of SOME/IP packets under attack conditions. It can be seen that their cosine similarity is lower than in attack-free conditions. This indirectly proves that the cosine similarity between the predicted and actual values possesses a certain capability to identify attacks and can be used as a rule for attack identification. Specifically, when an attacker injects malicious messages into the automotive Ethernet without knowledge of the target vehicle's Ethernet protocol, it will inevitably disrupt the temporal patterns of the data within the automotive Ethernet time window. As a result, the cosine similarity between the data in the window and the predicted values will decrease, enabling the detection of injected attacks. Additionally, for attackers who possess knowledge of the target vehicle's automotive Ethernet protocol, it would be extremely challenging to mimic the potential temporal patterns of the automotive Ethernet. Therefore, comparing the cosine similarity between the predicted results and

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the actual collected results can serve as one of the rules for automotive Ethernet intrusion detection, effectively identifying malicious intrusions that affect temporal patterns.

Apart from cosine similarity, other features may also serve as rules for detecting attacks, where the choice of rules will directly impact detection rates. In the future, we will assess the effectiveness of these rules in identifying various types of attacks. Due to limited research on intrusion detection in automotive Ethernet, existing studies have focused on different Ethernet protocols or attack types, which precludes direct comparison of experimental results. However, as research on automotive Ethernet IDS progresses, open-access real-world datasets may also become available, supporting the advancement and application of ICVs cyber security technologies.

V. CONCLUSIONS AND FUTURE WORK

This paper takes SOME/IP as an example and focuses on attack detection in automotive Ethernet. An intrusion detection method combining time series prediction and rule-based approaches is proposed. A prediction model that combines CNN-BiLSTM with attention mechanisms is utilized to learn the temporal patterns of data packets in automotive Ethernet. The cosine similarity between real messages and predicted results is employed as a rule to identify attacks. Experimental results demonstrate the effectiveness of the proposed method.

In the future, we plan to conduct experiments on real vehicle attacks and detection, as well as explore the deployment of the proposed model on in-vehicle devices. Additionally, we aim to identify more effective ways to establish rules for different attacks.

ACKNOWLEDGMENT

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